

Protocol for derivatization and determination of structural monosaccharides in crude fungal exopolysaccharide

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Abstract

Polysaccharides are composed of monomeric simple sugars, exhibiting substantial diversity for functional groups, found being suitable for vast biological activities in different living organisms. Difficulties in isolation, purification by professional methods and using costly equipment are the multiple challenges of characterizing the nature and properties of these important components of living cells. Several methods have been applied for the quantitative analysis of the carbohydrates. However, GC and GC-MS are the most appropriate techniques ever used to analyze complex structures of polysaccharide. Notably, so far, no commercial kit has been developed and offered, particularly for derivatization of polysaccharides. This study aims to confer a modified protocol for suitable derivatization of structurally unknown polysaccharides. The fungal exopolysaccharide from *Neopestalotiopsis* sp. has been used as the sample in this study. The study has established the multiple manual steps of the alditol acetate derivatization. The modifications have been highlighted as follows:

- Trifluoroacetic acid (TFA) has been used instead of conventional trichloroacetic acid for hydrolyzing the crude exopolysaccharide.
 - The standard sugars were treated separately, but not in the complex mixture.
 - The simplicity of the method, using ordinary instruments and short time accomplishment of procedures, collectively makes this technique as the most applicable in a common research laboratory where no more sophisticated equipment is available.
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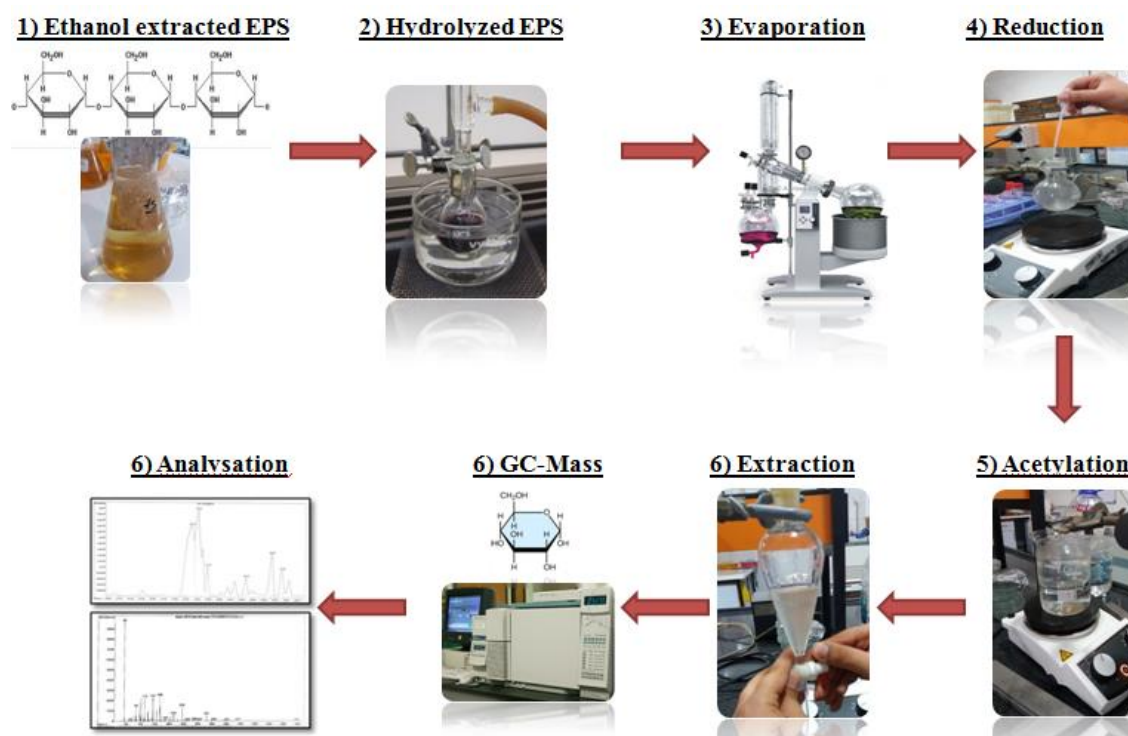
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Abbreviations:

GC: gas chromatography; MS: mass spectrometry; TFA: Trifluoroacetic acid; RT: room temperature; EPS: exopolysaccharide; D.W.: distilled water; hr: hours; M: molarity; PDA: potato dextrose agar; °C: Celsius; ml: milliliter; g/l: gram per liter, LB: luria bertani; RPM: revolutions per minute; min: minute;

GRAPHICAL ABSTRACT



Method details:

Isolation and cultivation of the microorganism

1. Preserve the collected biological sample (here, fruiting body of the fungus) at 4 °C in refrigerator.
2. Isolate fungus on PDA, incubate the petri plates at 28 °C for 7 days [1].

Note: Store mycelial culture of the isolate on Agar slants in test tubes in 4 °C and sub-culture monthly.

3. Set up the fermentation experiments in 250 ml Erlenmeyer flasks containing 50 ml semi-synthetic culture medium.

Note: The culture medium containing (g/l); glucose (30.0); yeast extract (6.0); polypeptone (2.0); $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (0.5); K_2HPO_4 (0.5) and $\text{MnSO}_4 \cdot 5\text{H}_2\text{O}$ (0.5) [2].

4. Punch out a piece of approximately 25 mm² of the agar medium with a sterile scalpel, homogenize by potter-type homogenizer and transfer it into the steam sterilized pre-culture medium containing (LB) broth.
5. Incubate at 30 °C on a rotary shaking incubator at 150 rpm for 2 days.
6. Transfer an aliquot of 10 ml from the pre-culture medium to inoculate 100 ml of semi-synthetic culture medium in 500 ml Erlenmeyer flasks.
7. Incubate the cultures at 30 °C on a rotary shaker, 150 rpm up to 7 days.

Crude exopolysaccharide extraction protocol

1. Collect a homogenous sample from shake flasks, centrifuge at 8000× g for 20 min.
2. Mix supernatant with four volumes of ethanol 96% (v/v) and keep overnight at 4 °C.
3. Centrifuge at 10,000× g for 20 min.
4. Collect the precipitate which is estimated as crude EPS.
5. Evaporate the excess ethanol from polysaccharide precipitates; air-dry overnight.
6. Adjust the pH of the raw EPS, 0.1% (w/v) aqueous solution, pH 3 with 10% trichloroacetic acid solution, and stir at 4 °C for 2 hr and then centrifuge for 30 min at 30,000× g.
8. Precipitate the supernatant with 4 volumes of absolute ethanol from the aqueous phase and resolve again in D.W.

9- Keep the creamy powder of EPS at 4 °C for further analyses.

Note: This procedure can repeat four times to purify the polysaccharide.

EPS derivatization protocol

Preparation of the stock solutions

1. Prepare 0.1 M Hydrochloric acid (HCl) stock solution: add 0.81 ml HCl (37%) to 25 ml D.W. firstly and bring to a final volume of 100 ml with D.W.
2. Prepare Ethanoic acid solution by mixing 4.3 ml of 0.1 M HCl stock solution with 4ml D.W. and bring to a final volume of 100 ml with absolute Methanol.
3. Make a solution of 15 ml Pyridine (anhydrous, 99.8 %, Sigma-Aldrich) in 30 ml D.W.
4. Prepare 2 M Trifluoroacetic acid (TFA) to a final volume of 100 ml in a volumetric flask using D.W.

Note: Pyridine is a toxic liquid and be careful when working with.

Note: Pyridine should be prepared as a daily solution and kept at -20 °C freezer 30 min before starting the experiments, but avoid freezing completely.

5. Prepare 1 N sulfuric acid using H₂SO₄ (98%) to a final volume of 100 ml using D.W.
6. Prepare 0.1 N sulfuric acid solution to a final volume of 100 ml using D.W.
7. Make a solution of sodium bicarbonate (NaHCO₃) (5%; w/v) using D.W.

Monosaccharides standard preparation

1. Prepare the pure monosaccharides (Sigma Aldrich), and dilute in D.W. with concentration of 0.01 M.

Note: Monosaccharides included (Glucose, Galactose and Mannose) [3-5], the standard procedure is followed according to the alditol derivatization protocol as below, except the hydrolyzation step.

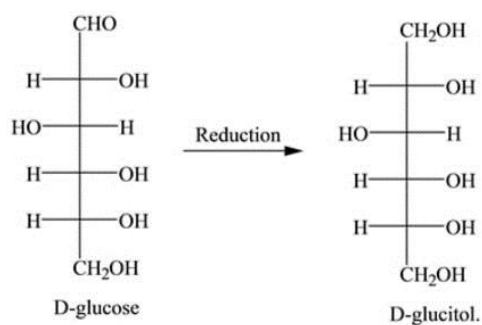
Alditol hydrolysis protocol

1. Add 15 mg of the crude extracted EPS with 10 ml 2 M TFA in an appropriate glass bottle, stir well using a magnetic stirrer.
2. Assemble a reflux apparatus using a 50 ml round-bottom flask and transfer the EPS solution to the flask.
3. Start hydrolysis reaction at 121 °C for 3 hr.
4. Evaporate the excess acid at 50 °C to obtain a viscous solution of carbohydrates (syrup) using an appropriate vacuum pump.
5. Dissolve the syrup in 5 ml D.W., stir well until the carbohydrate content is completely dissolved and keep it at 4 °C for further derivatization steps.

Derivatization details

1. Transfer the carbohydrates solution to a 250 ml round-bottom flask containing a magnet.
2. Clamp the bottle to ring stand at top of the magnet stirrer.
3. Add 40 mg Sodium borohydride (NaBH_4) to the solution slowly and allow stirring for 1hr at RT.

Note: Sodium borohydride (NaBH_4) is a mild reducing agent that is typically used to reduce aldehydes and ketones to their respective alcohols [6].

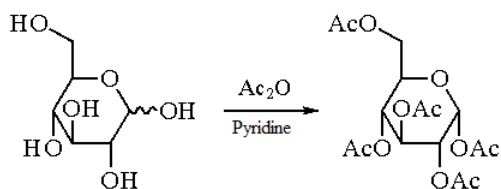


4. Add a few drops of Acetic acid to the solution to stop the reaction.
5. Allow the stirring to be continued for more time (20 min).
6. Evaporate the excess acid to obtain syrup.

7. Add 3 ml Ethanoic acid solution to the syrup under the fume hood; allow remaining for 1 min and evaporate the acid.

Note: This procedure (7) should be repeated for 3 times.

8. Add 5 ml acetic anhydride (Ac_2O) and 0.12 ml concentrated H_2SO_4 to the syrup; place it in hot water bath $80\text{ }^\circ\text{C}$ for 1 hr. Then, allow cooling at RT.
9. Add the previously prepared highly cold ($-20\text{ }^\circ\text{C}$) aqueous solution of pyridine and agitate on a magnet stirrer for 20 min. The acetylation step is completed in this section [7].



10. Transfer the solution to a 250 ml decanter funnel which clamped to ring stand under the fume hood.
11. Add the same volume of chloroform to the funnel, close the cap and reverse instantly for 3 times.
12. Two phases are appeared in the funnel; the upper phase includes pyridine and D.W. the bottom is the mixture of chloroform and derivatized saccharides.
13. Collect the sample mixed with chloroform from the funnel and discard the upper phase.
14. Rinse the funnel with the same volume of D.W., close the cap and reverse instantly for 3 times and collect the chloroform phase.
15. Repeat washing steps of the chloroform phase with 10 ml of H_2SO_4 1 N, 10 ml sodium bicarbonate (5%, w/v) and D.W. separately and one by one.
16. Dry the solution over 10 mg Sodium sulfate (Na_2SO_4). Filtrate the mixture to separate the salt and evaporate it under vacuum to obtain the syrup.
17. Dissolve the syrup in a few drops of chloroform.
18. Transfer immediately to a freezer and keep at $-20\text{ }^\circ\text{C}$ for further analysis.

GC-Mass analysis of alditol acetates protocol

1. Use this type of columns: HP-6890 coupled to HP-5973 mass selective detector (Agilent Technologies Co. Ltd., USA).

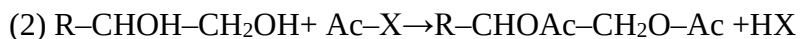
2. Perform all tests by using an auto sampler injector, and a 30 m × 0.25 mm × 0.25 μm HP-5 MS capillary column (Agilent Technologies).

3. The temperature program is adjusted as follow:

The oven temperature is initially set at 50°C, hold for 1 min and then it is increased at a rate of 5 °C/min to 100 °C, hold for 3 min and continued at 240 °C for 10 min at a rate of 10 °C/min and finished at 250 °C for 15 min. The auxiliary line and source temperatures are regulated at 285 °C. The MS is operated in electron impact (70 eV), using argon as a collision gas, at a pressure of 1 mTorr. Separation is performed from 250 °C to 285 °C at a rate of 1 °C/min. The volume injection of the samples is 1μl.

Method Validation

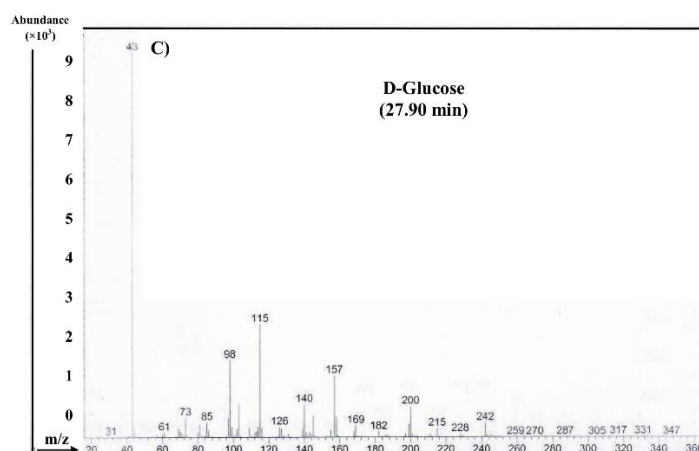
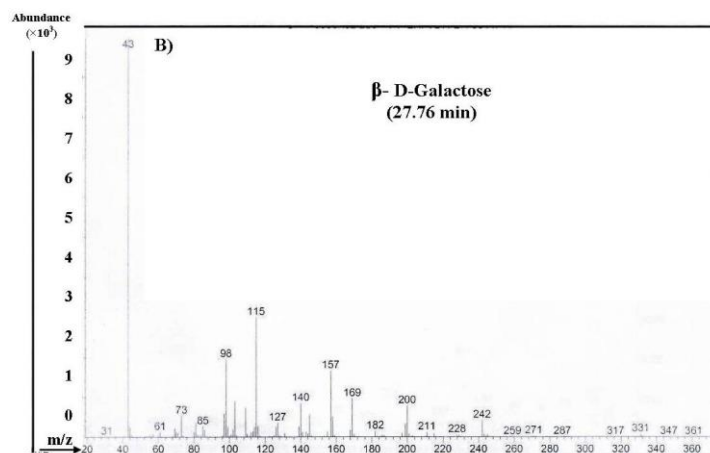
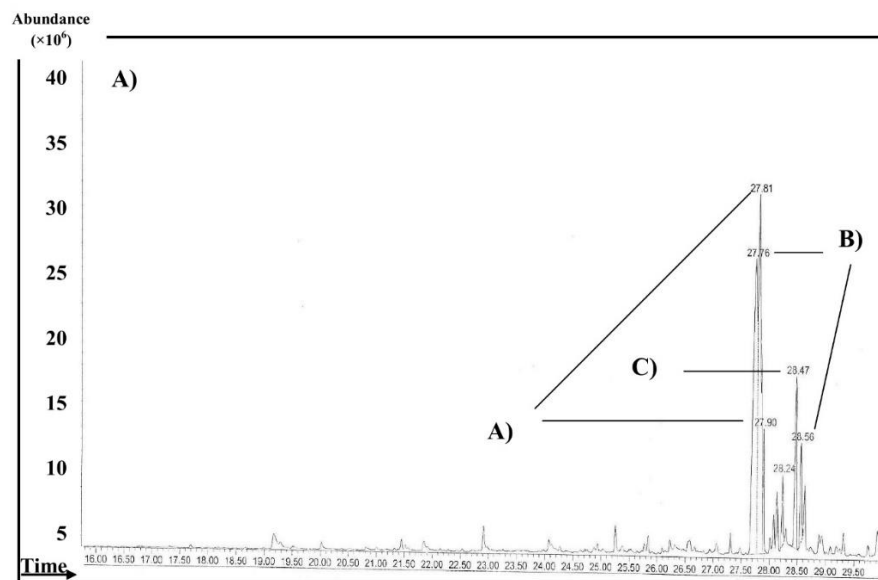
Different functional groups in carbohydrate structure exhibit different biological effects in living organisms. As carbohydrates often arise in complex form, a chromatographic method is necessary for their determination [7]. The most important methodology commonly used for carbohydrate analysis is GC and GC–MS techniques. These techniques offer several properties such as high sensitivity and resolution which are significant criteria for analysis of complex mixture [8]. However, the efficient derivatization procedure is required. Although, the problems such as the presence of high number of functional groups and different tautomeric forms make difficulties in GC-MS analysis of carbohydrate derivatives, but this technique is the best ever known. Derivatization of alditol acetate is appropriate method for analysis of such macromolecules that gives a single peak for each sugar by reducing the carbonyl groups of aldoses [9]. The derivatization reactions for alditol acetate are as follow:



In the present study characterization of the exopolysaccharide from *Neopestalotiopsis* sp. was performed based on hydrolysis with a subsequent derivatization of alditol acetate. Additionally, sugar carbonyl groups were reduced to alditols for decreasing the high number of peaks and then the remaining hydroxyl groups was acetylated. The partially methylated alditol acetates obtained in this assay can be directly analyzed by GC–MS.

Basically, characterization of the alditol acetate derivatives was performed by comparison of the m/z values and electron ionization of the obtained monosaccharide with those reported in the literatures. Consequently, comparison of the retention time of the separated derivatives with the standard sugars identified by the GC-Mass library (Wiley 2007) [10] support the previously results. Figure 1 shows the total peaks obtained by GC-MS analysis. Comparison of the retention times and mass-to-charge-ratio of the dervatized sugars obtained from EPS with those obtained from standard sugars treated with the same reactions demonstrated that D-glucose, D-galactose and D-glucitol were the most prominent components detected in the molecular structure of the EPS with the mass-to-charge-ratio of m/z 347, 361 and 375, respectively, based on GC-MS library.

As illustrated in Figure 1, the analyses of derivatized monosaccharides showed the retention time of D-glucose, β -D-galactose and D-glucitol with (27.90 and 27.81), (27.76 and 28.56) and 28.47, respectively. The three main components (D-glucose, β -D-galactose and D-glucitol) were similar to those in the GC-MS spectra of the hydrolyzed biopolymer [11, 12]. It was found that the molar ratio of β -D-galactose: D-glucose: D-glucitol was 59.86: 49.76: 12.62, respectively, when compared with the area of the peaks obtained from the standard sugars. These results proposed that the biopolymer produced by *Neopestalotiopsis* sp. is a polysaccharide containing β -D-galactose, D-glucose and D-glucitol with an approximate ratio of 5: 4: 1. Compositions of a number of fungal exopolysaccharides have been clarified and glucose appeared most often residue, while galactose and glucitol are seldom found as a component of the fungal EPS. In particular, concomitant presence of glucitol and galactose is also interesting, because they are rarely found both together among fungal EPS [13].



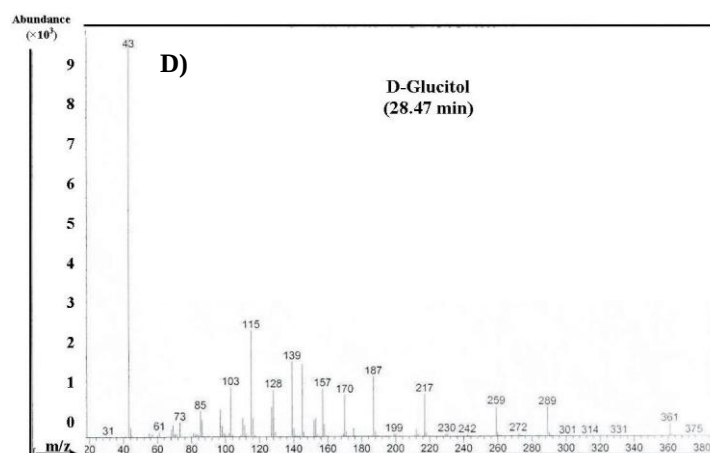


Fig.1. GC-MS spectra of the partially purified crude exopolysaccharide. The hydrolyzed and derivatized residues were analyzed. The peaks were identified by comparison of the spectra of related monosaccharides treated under the same conditions. Different components give separate peaks based on retention time (A). Retention time of 27.76 and 28.56 related to β -D-galactose with the m/z value of 361 (B), Retention time of 27.90 and 27.81 related to D-glucose with the m/z value of 347 (C) and retention time of 28.47 related to the D-glucitol with the m/z value of 375 (D).

Conflict of Interests

The author(s) have declared that there is not any conflict of interest.

Acknowledgement

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